

Web Dev + Database Roadmap

Web Dev + Database Roadmap — 3 Sessions/Week

Goal: Deployed website with a working database, submittable as a product **Timeline:** 13 weeks, 3 sessions/week (~1-1.5 hrs each), finals in November **Rule:** Fewer sessions, so don't skip one — missed sessions compress fast in this version.

Week 1 — How the Web Works + HTML Basics

- **Session 1:** Client-server model, URLs, HTTP/HTTPS, front-end vs back-end, hosting basics
- **Session 2:** HTML structure, common tags, semantic tags (header, nav, section, footer)
- **Session 3:** Forms + tables → start a static multi-page site
- **Checkpoint:** Static, unstyled multi-page HTML site

Week 2 — CSS + Bootstrap

- **Session 1:** Selectors, box model, flexbox
- **Session 2:** Grid, Bootstrap setup, grid system
- **Session 3:** Bootstrap components + responsive design → restyle Week 1 site
- **Checkpoint:** Styled, responsive site

Week 3 — Deploy

- **Session 1:** Choose free host (InfinityFree/000webhost), create account, explore file manager
- **Session 2:** Upload site, confirm public URL loads on another device
- **Session 3:** Check host's PHP version, fix any upload issues
- **Checkpoint:** Live site reachable by URL, no laptop needed

Week 4 — Database Design

(Overlaps DBMgt 1 — lean on that course here) - **Session 1:** Entities, attributes, relationships, keys, ERDs - **Session 2:** Normalization (1NF-3NF) - **Session 3:** SQL basics (CREATE TABLE, INSERT, SELECT, UPDATE, DELETE) → build your project schema in phpMyAdmin - **Checkpoint:** Live database schema on your host

Week 5 — PHP Basics

- **Session 1:** Variables, control flow, functions
- **Session 2:** Arrays, \$_GET/\$_POST
- **Session 3:** Sessions basics → deploy a test script (form → PHP → echo back)
- **Checkpoint:** Working PHP script live on host

Week 6 — Connecting PHP to the Database

- **Session 1:** PDO connection setup, running a SELECT query
- **Session 2:** Displaying DB results on a page, prepared statements
- **Session 3:** Inserting data via a form
- **Checkpoint:** Live page pulling and inserting real DB data

Week 7 — CRUD: Create + Read

- **Session 1:** Create: insert form → DB with validation
- **Session 2:** Read: list view of all records
- **Session 3:** Read: single-record detail view, test end-to-end

- **Checkpoint:** Create + Read live — **submittable minimum product**

Week 8 — CRUD: Update + Delete

- **Session 1:** Update: edit form pre-filled with existing data
- **Session 2:** Update: save changes back to DB
- **Session 3:** Delete with confirmation, full CRUD test pass
- **Checkpoint:** Full CRUD live and working

Week 9 — Auth + Related Tables

- **Session 1:** Registration with password_hash
- **Session 2:** Login with password_verify, sessions
- **Session 3:** Connect related tables (users → posts → comments), test full flow
- **Checkpoint:** Multi-table project with working login, live

Week 10 — JavaScript (Product-Focused Only)

- **Session 1:** Variables, functions, DOM basics
- **Session 2:** Events, form validation
- **Session 3:** Small dynamic UI touches → add directly to live product
- **Checkpoint:** Interactive touches added to live site

Week 11 — Polish + Fixes

- **Session 1:** UI cleanup, consistent styling pass
- **Session 2:** Error handling (empty forms, invalid input, DB errors)
- **Session 3:** Fix whatever broke earlier — something always does
- **Checkpoint:** Clean, stable, working product

Week 12 — Cold Testing + Buffer

- **Session 1:** Test entire site from a fresh browser (clear cookies) — simulate your prof's access
- **Session 2:** Fix issues found
- **Session 3:** Reserved slack — catch up if behind, deepen JS/PHP if ahead

- **Checkpoint:** Product survives a cold test

Week 13 — Finals Prep

- **Session 1-3:** Final review, walkthrough rehearsal, know your code well enough to explain any part
 - Light touch only — confidence, not new material
 - **Checkpoint:** Ready for finals
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Notes

- If a week runs long, pull from Week 12's buffer rather than compressing the next week.
- Weeks 7-9 are the core product — protect these if anything else slips.
- With only 3 sessions/week, a missed session matters more — if you have to skip, skip a polish week (10-11), not a core-build week (5-9).